5.1.3 Our place in the universse

	Learning outcomes	Additional guidance
(a)	*Describe and explain:*	
	(i) the use of radar-type measurements to determine distances within the solar system; how distance is measured and defined in units of time, assuming the relativistic principle of the invariance of the speed of light	HSW3, 11, 12
	(ii) effect of relativistic time dilation using the relativistic factor $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$	*M2.4*
	(iii) the measurement of relative velocities by radar observation	
	(iv) evidence for a 'hot big bang' origin of the universe from cosmological red-shifts (Hubble's law); cosmological microwave background.	HSW7,11 Development of the 'hot big bang' model
(b)	*Make appropriate use of, by sketching and interpreting:*	
	(i) logarithmic scales of magnitudes of quantities: distance, size, mass, energy, power, brightness.	*M2.5*
(c)	*Make calculations and estimates of:*	
	(i) distances and ages of astronomical objects	HSW7, 10
	(ii) distances and relative velocities from radar-type measurements.	HSW3